

Worksheet 3.1A

September 16, 2026

Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

1. Polynomial or not?

Consider

$$p(x) = 4 - 2x + 3x^4, \quad r(x) = \sqrt{x} + 2.$$

(a) For p , write standard form and identify its degree, leading coefficient, and constant term.

(b) Is r a polynomial function? Explain.

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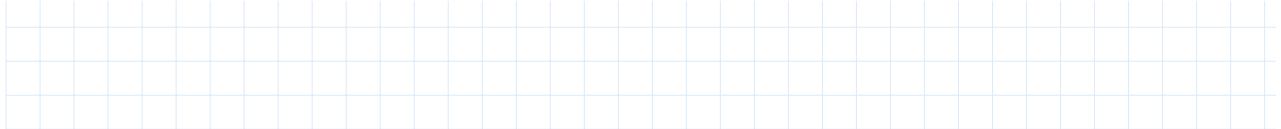
About 10 minutes total. Begin independently, then discuss. Show your work.

2. Zeros and end behavior

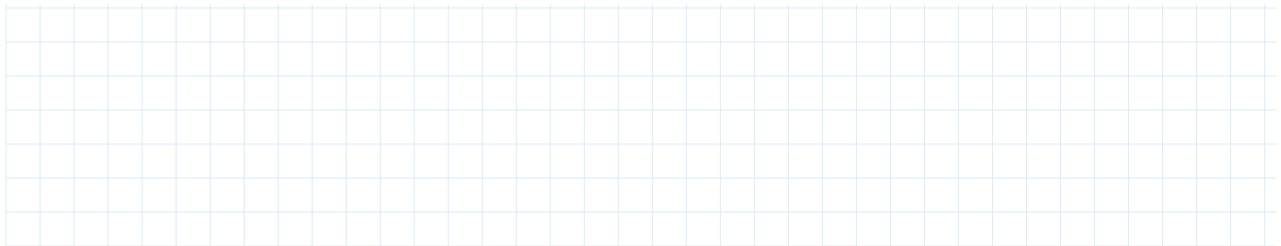
Let

$$P(x) = -(x + 2)(x - 1)^2.$$

(a) Find the degree and leading coefficient.



(b) Find each real zero and its multiplicity. State whether the graph crosses or touches the x -axis there.



(c) Write the limits of $P(x)$ as $x \rightarrow -\infty$ and $x \rightarrow \infty$.

